

The Laplace transform

Problem Sheet 2

1. Use the Laplace transform tables to find the Laplace transforms of each of the following functions. In each case specify the values of s for which the Laplace transform exists.

(a) $2e^{4t}$, (b) $3e^{-2t}$, (c) $5t - 3$, (d) $2t^2 - e^{-t}$, (e) $3\cos(5t)$.
(f) $10\sin(6t)$.

2. Express $F(t)$ in terms of the Heaviside step function $H(t - a)$ and hence find $\mathcal{L}\{F(t)\}$, where

$$F(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq 2, \\ 4 & \text{for } t > 2. \end{cases}$$

3. Use the First Shifting Theorem to find $\mathcal{L}\{f(t)\}$, where

$$f(t) = e^{\alpha t} \sinh(kt).$$

4. Use the table of Laplace transforms to invert the following:

(a) $\frac{1}{s+1}$, (b) $\frac{1}{(s-3)^2}$, (c) $\frac{a}{s(s+a)}$, (d) $\frac{k^2}{s(s^2+k^2)}$,
(e) $\frac{6s-4}{s^2-4s+20}$, (f) $\frac{e^{-2s}}{s^4}$.

5. Solve, using Laplace transforms:

(a)

$$\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 5y = 0, \quad \text{with } y(0) = 1, \quad y'(0) = -3.$$

(b)

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = e^{3t}, \quad \text{with } y(0) = 1, \quad y'(0) = 0.$$

(c)

$$\frac{dy}{dt} - 6y = 2\sin(3t), \quad \text{with } y(0) = -1.$$

6. Solve, using Laplace transforms:

$$\frac{d^2y}{dt^2} - 2a\frac{dy}{dt} + (a^2 + b^2)y = 0, \quad \text{with } y = 0, \quad \frac{dy}{dt} = 1, \quad \text{when } t = 0.$$

7. Solve, using Laplace transforms, the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = H(t) - H(t-4),$$

with the initial conditions $y(0) = 0$, $y'(0) = 0$.

8. Using the table of Laplace transforms, show that

$$(a) \mathcal{L}\{e^{-2t} \sin t\} = \frac{1}{s^2 + 4s + 5},$$

$$(b) \mathcal{L}\left\{\frac{1}{5} [1 - e^{-2t} \cos t - 2e^{-2t} \sin t]\right\} = \frac{1}{s(s^2 + 4s + 5)},$$

$$(c) \mathcal{L}\left\{\frac{1}{5} [1 - e^{-2v} \cos v - 2e^{-2v} \sin v] H(v)\right\} = \frac{e^{-3s}}{s(s^2 + 4s + 5)},$$

where $v = t - 3$.

Use Laplace transform techniques to solve the ordinary differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = H(t-3),$$

with $y(0) = 0$ and $y'(0) = 1$.

9. Find the Laplace transform of the function

$$f(t) = 1 - e^{-t}$$

Hence, verify that the final value theorem applies.

10. Use the definition of the Laplace transform of a function,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt,$$

and integration by parts to show the following.

$$(a) \mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}.$$

$$(b) \mathcal{L}\{\sinh(at)\} = \frac{a}{s^2 - a^2}.$$

$$(c) \mathcal{L}\{\cosh(at)\} = \frac{s}{s^2 - a^2}.$$

Hint:

$$\frac{d}{dt}(\sinh(at)) = a \cosh(at), \quad \frac{d}{dt}(\cosh(at)) = a \sinh(at).$$

$$(d) \text{ [HARD] } \mathcal{L}\{t \sin(kt)\} = \frac{2ks}{(s^2 + k^2)^2}.$$

$$(e) \text{ [HARD] } \mathcal{L}\{t \cos(kt)\} = \frac{s^2 - k^2}{(s^2 + k^2)^2}.$$

Prerequisites:

$P(n) = \{\text{questions with prerequisite Parts } \leq n\}$

$A = \text{all of PS2.}$

$P(5) = \{1, 4a, 4c^*, 4d^*, 10\}$

$P(6) = \{1, 2, 4a, 4c^*, 4d^*, 5, 6, 10\} = A \setminus \{3, 4b, 4e, 4f, 7, 8, 9\}$, * = using partial fractions

$P(7) = A$